Demystifying the channels

The nomenclature 2.1, 4.1, and 5.1 denotes the number of channels of sound that a speaker can reproduce. Simply put, the number of satellites is represented by the figure before the dot, and the subwoofer—which is almost always 1—after it.

A 2.1 speaker set will have two sound channels, and two satellites and a woofer. A 4.1 set similarly implies four speakers (two front and two rear) and a woofer. The term 5.1 denotes a 6-channel setup with five satellites (two front, two rear and a centre) and a woofer. Similarly, a 7.1 system will have seven satellites and one woofer. There are also systems that have two woofers; in case of a 6-channel setup, this would be shown as a 5.2 setup. However, such systems for computers are extremely rare.

actually looked like a slim DVD player at first glance, and we did a double take just to be sure.

Two 5.1 speakers sported tower speakers—the Genius GHT-S200 with a silver and black finish, and the Zebronics ZEB-SW12000R in a complete metallic silver and chrome finish. Incidentally, these two models sported tall subwoofers that were sized on par with a mid-sized computer cabinet.

Altec Lansing's GT-5051R surprised us when we found only three satellites amongst the packing. We were surprised to see that this model integrates the rear speakers with the front speakers itself, but with a different direction of fire. Altec Lansing calls this "Side Firing Dipole technology." Just three satellites for a 5.1 setup is something unique—space-conscious users, take note!

Sounding Off

We fired up the *Half-life 2* demo to check out the gaming performance of these beauties. The Artis S6600R was a clear winner here, surprisingly, considering its overall power. Its bass reproduction was good, and the voice commands of the squad could be easily heard above all the gunfire and explosions. The Philips HTR5000 was a distant second while the Logitech Z-5500D was close on its heels at third place.

The Frontech JIL 1875 and the Zebronics ZEB-SW12000R brought up the rear with the lowest scores in the gaming segment.

Coming to the DVD tests, the Philips HTR5000 topped both the tests with clear sound and hardly any discernable distortion. The bass was flat and the highs and ambient background sounds faithfully reproduced. The Artis S6600R came in second, with the second-best overall scores.

Also worthy of a mention are the Creative Inspire GD580, Philips MMS 5.500 i/C and the Logitech Z-5500D which produced some decent scores. The GD 580 basically lacked the bass punch necessary, owing to its lower RMS rating. The Z-5500D was quite the opposite: the 187-watt RMS subwoofer overpowered everything else and the explosions in the movie rocked us, but the bass tended towards rumbling and wasn't as tight as we'd have liked it to be.

The lowest scorers in this segment were for the Frontech JIL-1875 and the Intex IT-4900W. There was a lot of distortion, especially in the high-pitched sounds. Bass was also below par and explosions sounded flat. Not recommended for DVD viewing!

The DVD test track was tested for treble, vocals and bass, and the undisputed winner was the Logitech Z-5500D. This speaker set left the others biting the dust. The closest rivals were the two Philips siblings the HTR5000 and the MMS 5.500 i/C, in that order.

The Frontech JIL-1875 sounded pathetic in the DVD audio test, with some of the lowest scores ever recorded. The Altec Lansing GT-5051R also could not handle this test and posted some low scores. A surprise bad performer here was the Creative Inspire GD580; it just didn't impress.

Next, we decided to stress-test the 5.1s with our frequency tests. These tests stress the speakers because they contain sounds of a particular frequency played continuously.

The 50 Hz test was won by the the Logitech Z-5500D with a near-

Philips MMS 5.500 i/C

4.1—A Dying breed

This time round we received just three 4.1 speakers to test. Certain large vendors such as Logitech and Altec Lansing have done away with the 4.1 product line almost entirely. There is a strong technical reason for this. Let us explain.

4.1 speakers are of two types—true 4.1 and pseudo 4.1. A true 4.1 speaker set has four channels and a subwoofer that derives low frequency sounds from the front two channels. Pseudo 4.1s have four satellites but actually have just two channels, and the rear speakers and woofer do not have discrete sound channels. A Pseudo 4.1 is therefore nothing more than a 2.1 with two extra speakers. In movies supporting Dolby surround, a 4.1 will not be able to deliver accurate positional sound.

Coming to the gaming segment where EAX rules, 4.1 is once again a big no-no.

A 4.1 is therefore not suitable for either the music or gaming/DVD segment, which is better served by 2.1 and 5.1 systems respectively.

For those of you still hanging on to that old SoundBlaster 4.1 channel sound card, we tested the three 4.1s (two sets out of these were "Pseudo sets"). The Creative Inspire 4400 surprisingly lost out badly and was placed last. Sadly this was the only "true 4-channel speakers" of the lot. The winner among these bad boys (pun intended) was the Frontech JIL 1867. The other contender the Intex IT 2880W was marginally better in *Half Life 2*.

For music playback, the Frontech JIL 1867 came out tops. In case you absolutely must have a 4.1 for whatever reason, this is your best bet.

For an overall picture, it must be said that the 4.1s are worse than the smaller 2.1 sets, and don't even come close to 5.1 territory.

4.1 sets were originally meant to be a cheap replacement to the discrete six-channel 5.1s, which were costly up until a few years ago. Today, the cost of an entry level 5.1 speaker set is just a couple of thousand rupees more than a similar 4.1 set. Anyone wanting a surround sound experience is, in our opinion, better served by investing a bit more in a 5.1 system. We feel the number of 4.1s we received as compared to the other categories speak volumes about good old "Pseudo Surround." R.I.P.!